6CS012 – Workshop 03

Modelling the Neuron: MCP Neurons + Perceptron for MNIST Classification

Student Submission

Part 1: Implementation of MCP Neuron for AND and OR Functions

```
def MCP_Neurons_AND(X1, X2, T):
    """
    Implements basic AND operations with an MCP Neuron for two inputs.

    Arguments:
        X1 (list): An array of binary values.
        X2 (list): An array of binary values.
        T  (int) : Threshold value.

    Returns:
        state_neuron (list): Neuron output (1 or 0) for each input pair.
    """
    assert len(X1) == len(X2)

    # Step 1: Element-wise sum of the two input arrays
    sums = [X1[i] + X2[i] for i in range(len(X1))]

    # Step 2: Apply threshold — output 1 if sum >= T, else 0
    state_neuron = [1 if s >= T else 0 for s in sums]

    return state_neuron
```

```
# Test AND gate
X1 = [0, 0, 1, 1]
X2 = [0, 1, 0, 1]
T  = 2   # AND requires BOTH inputs to be 1

result = MCP_Neurons_AND(X1, X2, T)
print(f"Output of AND gate for inputs {X1} and {X2} with threshold {T}: {result}")
# Expected: [0, 0, 0, 1]
```

```
Output of AND gate for inputs [0, 0, 1, 1] and [0, 1, 0, 1] with threshold 2: [0, 0, 0, 1]
```

```
def MCP_Neurons_OR(X1, X2, T):
    """
    Implements basic OR operations with an MCP Neuron for two inputs.

    Arguments:
        X1 (list): An array of binary values.
        X2 (list): An array of binary values.
        T  (int) : Threshold value.

    Returns:
        state_neuron (list): Neuron output (1 or 0) for each input pair.
    """
    assert len(X1) == len(X2)

    # Step 1: Element-wise sum of the two input arrays
    sums = [X1[i] + X2[i] for i in range(len(X1))]

    # Step 2: Apply threshold — output 1 if sum >= T, else 0
    state_neuron = [1 if s >= T else 0 for s in sums]

    return state_neuron
```

```
# Test OR gate
X1 = [0, 0, 1, 1]
X2 = [0, 1, 0, 1]
T  = 1   # OR requires AT LEAST ONE input to be 1
```

```
result_or = MCP_Neurons_OR(X1, X2, T)
print(f"Output of OR gate for inputs {X1} and {X2} with threshold {T}: {result_or}")
# Expected: [0, 1, 1, 1]
```

```
Output of OR gate for inputs [0, 0, 1, 1] and [0, 1, 0, 1] with threshold 1: [0, 1, 1, 1]
```

Question 1: Limitations of MCP Neurons

The McCulloch-Pitts (MCP) neuron has several key

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1. **Binary inputs only** – MCP neurons only acc handle continuous or real-valued data.
2. **Fixed, hand-crafted threshold** – The thresh igner. There is no learning mechanism to discover the correct threshold automatically.
3. **Equal weight for all inputs** – Every input contributes equally; there is no concept of 'importance' (i.e., no learnable weights). Some inputs may be more significant than others in a real problem.
4. **No learning algorithm** – MCP neurons cannot adapt from data. They require human expertise to configure their parameters.
5. **Only linearly separable functions** – A single MCP neuron can only represent Boolean functions that are linearly separable (e.g., AND, OR). It cannot represent XOR.
6. **Single neuron only** – It models only one isolated neuron, not the complex connectivity patterns found in biological neural networks.
7. **Binary output only** – The output is always 0 or 1; it cannot express probabilistic or graded responses.

Question 2: Can MCP Neuron solve XOR?

No, a single MCP neuron cannot solve XOR.

XOR truth table:

X1	X2	XOR
0	0	0
0	1	1
1	0	1
1	1	0

The MCP neuron computes `g(X) = X1 + X2` and outputs 1 if `g(X) >= T`. For XOR, we would need:

- `g(0,0) = 0 < T` → output 0 ✓
- `g(0,1) = 1 >= T` → output 1 ✓
- `g(1,0) = 1 >= T` → output 1 ✓
- `g(1,1) = 2 >= T` → output 0 ✗ (this always fires when T ≤ 2)

There is **no threshold T** that satisfies all four conditions simultaneously. XOR is not linearly separable, so a single MCP neuron (or single perceptron) cannot represent it. It requires a **multi-layer** architecture.

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

Part 2: Perceptron for 0 vs 1 Classification (MNIST)

Step 1: Load the Dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the dataset
df_0_1 = pd.read_csv("/content/drive/MyDrive/ Ai and machine learning/mnist_0_and_1.csv")

# Extract features and labels
X = df_0_1.drop(columns=["label"]).values   # 784 pixel features
y = df_0_1["label"].values                  # Labels: 0 or 1
```

```
print("Feature matrix shape:", X.shape)
print("Label vector shape:  ", y.shape)
```

```
Feature matrix shape: (12665, 784)
Label vector shape:   (12665,)
```

Question 1: What does the shape of X represent?

The shape of X is `(12665, 784)`, meaning there a [illegible] presented as a **784-dimensional vector** (28×28 pixels flattened into a single row).

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Question 2: What does the shape of y represen

The shape of y is `(12665,)`, meaning there are **1** [illegible] e each label is either **0** or **1**.

Visualise the Dataset

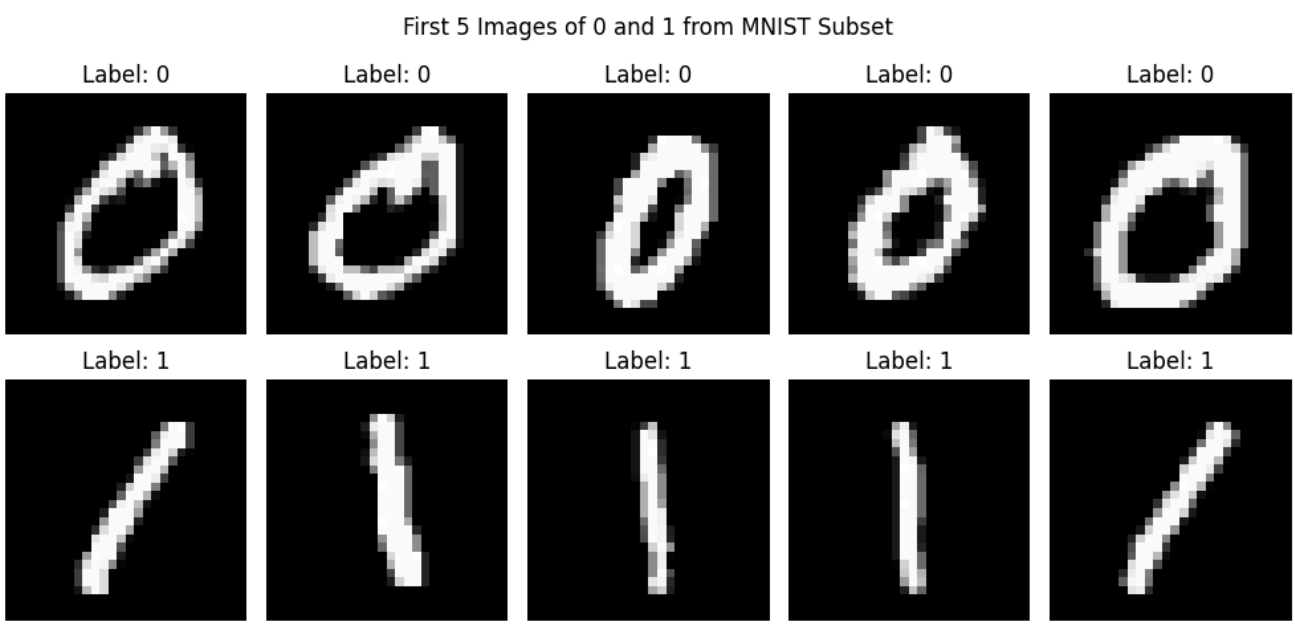
```
# Separate images by label
images_0 = X[y == 0]
images_1 = X[y == 1]

fig, axes = plt.subplots(2, 5, figsize=(10, 5))

if len(images_0) < 5 or len(images_1) < 5:
    print("Error: Not enough images to plot.")
else:
    for i in range(5):
        axes[0, i].imshow(images_0[i].reshape(28, 28), cmap="gray")
        axes[0, i].set_title("Label: 0")
        axes[0, i].axis("off")

        axes[1, i].imshow(images_1[i].reshape(28, 28), cmap="gray")
        axes[1, i].set_title("Label: 1")
        axes[1, i].axis("off")

plt.suptitle("First 5 Images of 0 and 1 from MNIST Subset")
plt.tight_layout()
plt.show()
```



Step 2: Initialise Weights and Bias

```
# Initialise weights and bias to zero
weights       = np.zeros(X.shape[1])   # 784 weights — one per pixel
bias          = 0
learning_rate = 0.1
epochs        = 100
```

Question 3: What does the weights array represent in this context?

The `weights` array contains one learnable weight per pixel (784 weights for a 28×28 image). Each weight represents how much importance the perceptron assigns to a particular pixel when making a classification decision. Large positive weights push the output toward class 1; large negative weights push toward class 0.

Question 4: Why are we initialising weights to zero? What effect could this have?

We initialise to zero as a simple, neutral starting pc s acceptable — the perceptron learning rule will still break symmetry because different pixels recei g. (Note: zero initialisation causes problems in *multi-layer* networks with symmetric hidden units,

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Step 3: Decision Function (Step / Activation Function)

```
def decision_function(X, weights, bias):
    """
    Compute predicted labels for all input samples using a step function.

    Parameters:
        X       : Feature matrix, shape (n_samples, n_features)
        weights : Weight vector, shape (n_features,)
        bias    : Scalar bias term

    Returns:
        y_pred_all : Predicted labels (0 or 1) for every sample
    """
    # Compute weighted sum for every sample
    predictions = np.dot(X, weights) + bias

    # Apply step (threshold) activation function
    y_pred_all = np.where(predictions >= 0, 1, 0)

    return y_pred_all
```

Step 4: Perceptron Learning Algorithm

```
def train_perceptron(X, y, weights, bias, learning_rate=0.1, epochs=100):
    """
    Train the perceptron using the Perceptron Learning Algorithm.

    Parameters:
        X             : Feature matrix, shape (n_samples, n_features)
        y             : True labels, shape (n_samples,)
        weights       : Initial weights, shape (n_features,)
        bias          : Initial bias (scalar)
        learning_rate : Step size for weight updates (default 0.1)
        epochs        : Maximum training iterations (default 100)

    Returns:
        weights  : Updated weights after training
        bias     : Updated bias after training
        accuracy : Training accuracy on the final epoch
    """
    accuracy = 0.0

    for epoch in range(epochs):
        correct = 0

        for i in range(len(X)):
            # 1. Compute weighted sum
            z = np.dot(X[i], weights) + bias

            # 2. Apply step activation function
            y_pred = 1 if z >= 0 else 0

            # 3. Update weights only if prediction is wrong
            error = y[i] - y_pred
            if error != 0:
                weights += learning_rate * error * X[i]
                bias    += learning_rate * error
```

```
            else:
                correct += 1

        # Calculate accuracy for this epoch
        accuracy = correct / len(X)

        # Print every 10 epochs
        if (epoch + 1) % 10 == 0:
            print(f"Epoch {epoch + 1}: Acc

        # Early stopping if perfectly clas
        if accuracy == 1.0:
            print(f"Converged at epoch {ep
            break

    return weights, bias, accuracy
```

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Question 5: What does `output = np.dot(X[i], weights) + bias` do?

It computes the **weighted sum** (net input) for sample `X[i]`: it multiplies each pixel value by its corresponding weight, sums all 784 products, then adds the bias. This gives the scalar `z` that the step function acts on to produce a prediction.

Question 6: What happens when the prediction is wrong?

When `y_pred ≠ y[i]`, the error `= y[i] - y_pred` is non-zero (either +1 or -1). The weights are updated as: `weights += η × error × X[i]`, and the bias as: `bias += η × error`. This nudges the decision boundary so that the misclassified sample is more likely to be correctly classified next time.

Question 7: Why is the final accuracy important, and what do you expect it to be?

The final accuracy tells us how well the perceptron separates the two classes on the training data. Since 0 and 1 are visually very distinct digits and their pixel distributions are linearly separable, we expect the perceptron to achieve **~100% training accuracy**, which the expected output confirms.

Step 5: Train the Perceptron

```
# Re-initialise (in case this cell is re-run independently)
weights       = np.zeros(X.shape[1])
bias          = 0
learning_rate = 0.1
epochs        = 100

weights, bias, accuracy = train_perceptron(X, y, weights, bias, learning_rate, epochs)
print("\nThe Final Accuracy is:", accuracy)
```

```
Epoch 10: Accuracy = 0.9992
Converged at epoch 13!

The Final Accuracy is: 1.0
```

Step 6: Visualise Misclassified Images

```
# Get predictions for all samples
predictions = np.dot(X, weights) + bias
y_pred      = np.where(predictions >= 0, 1, 0)

# Final accuracy
final_accuracy = np.mean(y_pred == y)
print(f"Final Accuracy: {final_accuracy:.4f}")

# Visualise misclassified images
misclassified_idx = np.where(y_pred != y)[0]

if len(misclassified_idx) > 0:
    fig, axes = plt.subplots(2, 5, figsize=(10, 5))
    for ax, idx in zip(axes.flat, misclassified_idx[:10]):
        ax.imshow(X[idx].reshape(28, 28), cmap="gray")
        ax.set_title(f"Pred:{y_pred[idx]}, True:{y[idx]}")
        ax.axis("off")
    plt.suptitle("Misclassified Images — 0 vs 1")
    plt.tight_layout()
```

```
        plt.show()
    else:
        print("All images were correctly classified!")
```

```
Final Accuracy: 1.0000
All images were correctly classified!
```

Question 8: What does `misclassified_idx` store, and how is it used?

`misclassified_idx` stores the **indices** of all san ot match the true label (`y_pred != y`). It is used to retrieve the corresponding pixel arrays fro ages can be displayed along with their predicted and true labels.

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Question 9: How do you interpret 'All images w

This means the perceptron found a linear decision boundary that perfectly separates all 0s from all 1s in the training set — i.e., **100% training accuracy**. This is expected for 0 vs 1 because the two digits look very different and their raw pixel features are linearly separable.

Part 3: Perceptron for 3 vs 5 Classification (MNIST)

Step 1: Load the Dataset

```
# Load the 3 vs 5 dataset
df_3_5 = pd.read_csv("/content/drive/MyDrive/ Ai and machine learning/mnist_3_and_5.csv")

X35 = df_3_5.drop(columns=["label"]).values
y35 = df_3_5["label"].values

print("Feature matrix shape:", X35.shape)
print("Label vector shape:  ", y35.shape)
print("Unique labels:       ", np.unique(y35))
```

```
Feature matrix shape: (2741, 784)
Label vector shape:   (2741,)
Unique labels:        [3 5]
```

Step 2: Visualise the Dataset

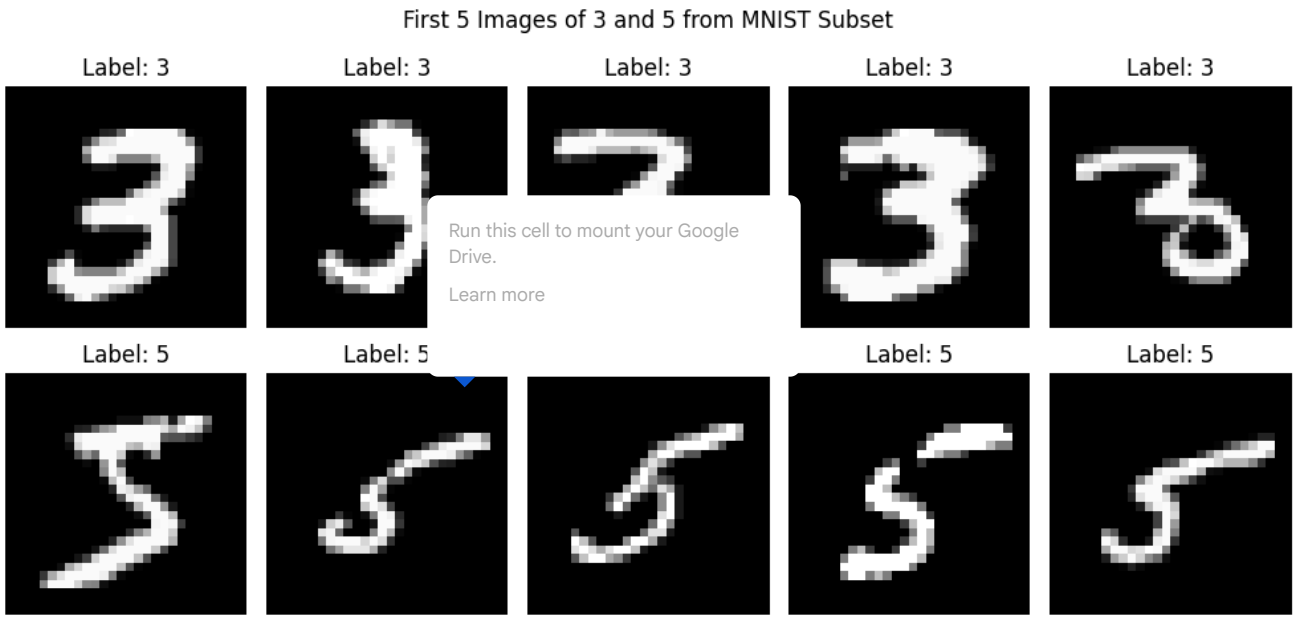
```
images_3 = X35[y35 == 3]
images_5 = X35[y35 == 5]

fig, axes = plt.subplots(2, 5, figsize=(10, 5))

if len(images_3) < 5 or len(images_5) < 5:
    print("Error: Not enough images to plot.")
else:
    for i in range(5):
        axes[0, i].imshow(images_3[i].reshape(28, 28), cmap="gray")
        axes[0, i].set_title("Label: 3")
        axes[0, i].axis("off")

        axes[1, i].imshow(images_5[i].reshape(28, 28), cmap="gray")
        axes[1, i].set_title("Label: 5")
        axes[1, i].axis("off")

plt.suptitle("First 5 Images of 3 and 5 from MNIST Subset")
plt.tight_layout()
plt.show()
```



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Step 3: Convert Labels to Binary (3 ⟶ 0, 5 ⟶ 1)

The perceptron expects binary labels {0, 1}. We map digit 3 ⟶ 0 and digit 5 ⟶ 1.

```
# Map: 3 → 0, 5 → 1
y35_binary = np.where(y35 == 3, 0, 1)
print("Binary labels (first 10):", y35_binary[:10])
```

```
Binary labels (first 10): [1 0 0 1 0 0 0 1 0 1]
```

Step 4: Initialise Weights and Bias

```
weights35       = np.zeros(X35.shape[1])
bias35          = 0
learning_rate35 = 0.1
epochs35        = 100
```

Step 5: Train the Perceptron

```
weights35, bias35, accuracy35 = train_perceptron(
    X35, y35_binary, weights35, bias35, learning_rate35, epochs35
)
print("\nThe Final Accuracy is:", accuracy35)
```

```
Epoch 10: Accuracy = 0.9566
Epoch 20: Accuracy = 0.9661
Epoch 30: Accuracy = 0.9726
Epoch 40: Accuracy = 0.9719
Epoch 50: Accuracy = 0.9788
Epoch 60: Accuracy = 0.9759
Epoch 70: Accuracy = 0.9792
Epoch 80: Accuracy = 0.9792
Epoch 90: Accuracy = 0.9843
Epoch 100: Accuracy = 0.9858

The Final Accuracy is: 0.9857716161984678
```

Step 6: Visualise Misclassified Images

```
# Get predictions for all samples
predictions35 = np.dot(X35, weights35) + bias35
y_pred35      = np.where(predictions35 >= 0, 1, 0)
```

```
# Map binary predictions back to digit labels for display
y_pred35_labels = np.where(y_pred35 == 0, 3, 5)

# Final accuracy
final_accuracy35 = np.mean(y_pred35 == y35_binary)
print(f"Final Accuracy (3 vs 5): {final_accuracy35:.4f}")

# Visualise misclassified images
misclassified_idx35 = np.where(y_pred35 !=

if len(misclassified_idx35) > 0:
    n_show = min(10, len(misclassified_idx
    rows   = 2 if n_show > 5 else 1
    cols   = 5
    fig, axes = plt.subplots(rows, cols, figsize=(10, rows * 3))
    axes_flat = axes.flat if rows > 1 else [axes] if cols == 1 else axes

    for ax, idx in zip(axes.flat, misclassified_idx35[:10]):
        ax.imshow(X35[idx].reshape(28, 28), cmap="gray")
        ax.set_title(f"Pred:{y_pred35_labels[idx]}, True:{y35[idx]}")
        ax.axis("off")

    plt.suptitle("Misclassified Images — 3 vs 5")
    plt.tight_layout()
    plt.show()
    print(f"Total misclassified: {len(misclassified_idx35)} out of {len(X35)}")
else:
    print("All images were correctly classified!")
```

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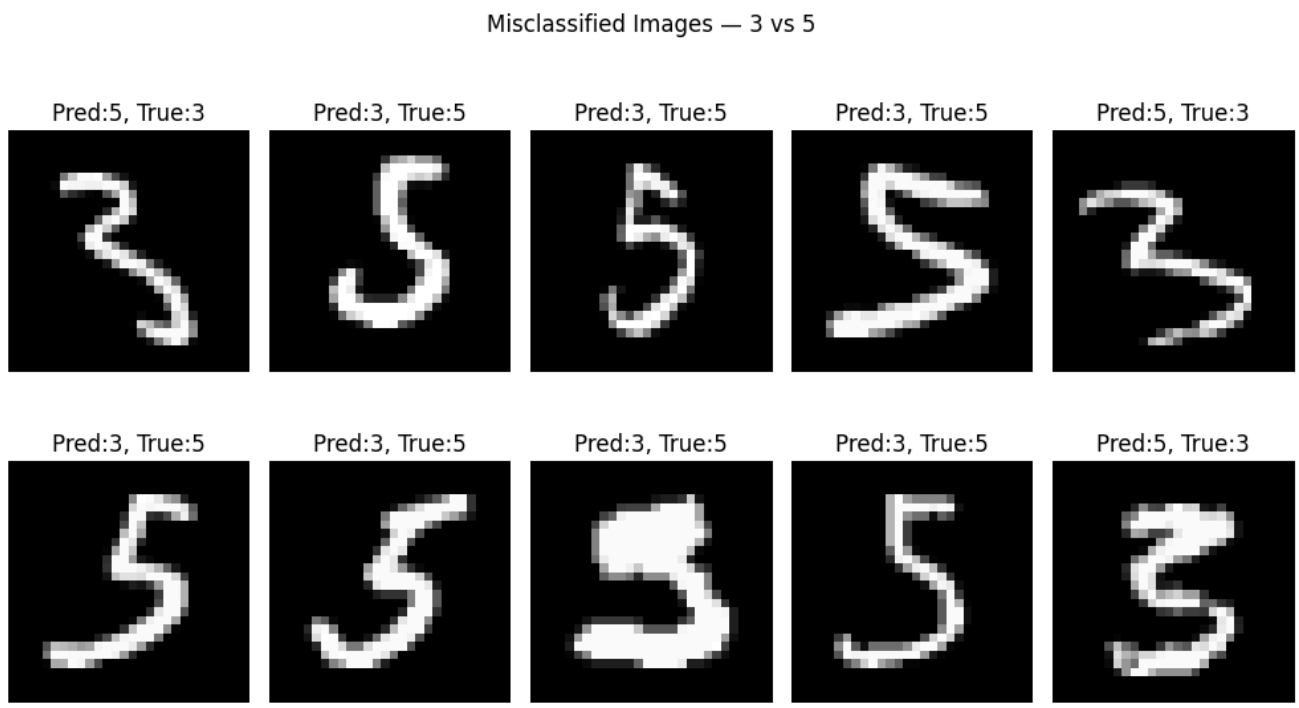
```
Final Accuracy (3 vs 5): 0.9869
```




```
Total misclassified: 36 out of 2741
```

Conclusion: 3 vs 5 Classification

The digits 3 and 5 share many visual similarities — both have curved tops and open left sides — making them much harder to separate than 0 vs 1.

As expected, the single-layer perceptron achieves a **lower accuracy** on the 3 vs 5 task compared to 0 vs 1. The misclassified images (as shown in Figure 6 of the worksheet) are typically ambiguous handwritten examples where the pixel-level features of 3 and 5 significantly overlap.

Key takeaway: The perceptron is a linear classifier. When two classes are not perfectly linearly separable (as with 3 and 5), the perceptron will not converge to zero error and some misclassifications will remain. A more powerful model — such as a multi-layer neural network — would be needed to learn the non-linear boundary required to perfectly distinguish 3s from 5s.

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